A Mini Project Report

on

BOOK MY DOCTOR

Submitted in partial fulfilment of the requirement of

Bachelor of Technology (S.Y)

in

COMPUTER ENGINEERING

submitted by

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Approval Sheet

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BOOK MY DOCTOR

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Declaration

We declare that this written submission for S.Y. Mini Project entitled **"Book My Doctor"** represent our ideas in our own words and where others' ideas or words have been included. We have adequately cited and referenced the original sources. We also declared that we have adhere to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any ideas / data / fact / source in our submission. We understand that any violation of the above will cause for disciplinary action by institute and evoke penal action from the sources which have thus not been properly cited or from whom paper permission have not been taken when needed.

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ABSTRACT

Healthcare systems worldwide are rapidly adopting digital solutions to make medical services more accessible, reliable, and efficient. Traditional appointment booking methods often involve long queues, manual scheduling, and limited access to specialized doctors, especially for patients in rural or remote areas. These challenges result in delays, inconvenience, and reduced quality of care.

To address these issues, our project**"Book My Doctor"** presents a web-based healthcare platform that connects patients with doctors in a seamless and user-friendly manner. The system allows patients to search doctors by specialty, and availability, and book appointments online. It also integrates teleconsultation services to ensure continuous medical care without the need for physical visits. Key features of the system include **AI-powered chatbot support for instant assistance, home visit booking options for elderly** and **disabled patients,** and **digital health record management** with e-prescriptions and reminders. These features reduce patient waiting time, improve healthcare accessibility, and enhance doctor efficiency by automating scheduling and record-keeping.

The proposed platform ensures data privacy and security through encrypted storage, while maintaining a responsive and intuitive interface.By combining telemedicine, automation, and secure digital health management, Book My Doctor promotes smarter, faster, and more reliable healthcare services.

Keywords: Online Appointment System, Telemedicine, Digital Healthcare, AI Chatbot, Health Records Management, Patient-Centric Platform, Web-Based System, Smart Healthcare Services.

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Chapter 1

Introduction

1.1 Background

Healthcare has always been a fundamental necessity, but the methods of accessing it have traditionally been inconvenient and inefficient. In the past, patients were required to physically visit hospitals or clinics to schedule appointments, which often resulted in long waiting times, overcrowding, and delays in receiving medical attention. These traditional methods were highly dependent on manual scheduling, which increased the chances of errors and further inconvenienced both patients and doctors.

With the rapid growth of the internet and smartphones, digital transformation began to reshape multiple industries such as shopping, banking, and education. Healthcare also started adopting digital solutions to improve accessibility and efficiency.The introduction of **online doctor appointment** platforms provided patients with an easy way to search for doctors, view their qualifications, compare expertise, and book consultations remotely. Such platforms also integrated **telemedicine**, enabling patients to consult doctors online through video calls and chat, which became especially crucial during the COVID-19 pandemic when physical interactions were restricted.

Despite these advancements, many existing platforms still face limitations, such as lack of home visit services for elderly and disabled patients, limited coverage in rural areas, poor doctor–patient commu-

nication outside clinic hours, and inadequate integration of smart technologies like AI chatbots for instant guidance. These gaps highlight the need for a more **comprehensive, reliable, and patient-centric digital healthcare system.**

The Book My Doctor platform is designed to address these shortcomings by combining traditional online booking with enhanced features such as **teleconsultation**, **home visit scheduling**, **AI-powered chatbot assistance**, **digital health record management**, and automated reminders. By leveraging modern web technologies and secure databases, the platform aims to provide **convenient, quick, and reliable access to healthcare** for patients, while helping doctors manage their appointments more efficiently.

1.2 Motivation

The motivation behind developing the **Book My Doctor** platform arises from the shortcomings of traditional healthcare systems and the limitations of existing digital solutions. Patients often face long queues, manual scheduling errors, and wasted time when booking appointments through offline methods. This creates frustration and reduces the efficiency of healthcare delivery.

Certain groups of patients, such as the elderly and those with chronic illnesses, often require **home visit services** as they are unable to travel easily to clinics or hospitals. However, most existing platforms lack this critical feature, limiting healthcare accessibility for vulnerable populations. Furthermore, the absence of **smart assistance tools**, such as chatbots, creates difficulty for patients who need quick answers to common questions or guidance during the booking process.

There is also a **persistent communication gap between doctors**

and patients, since most interaction ends after the physical consultation, leaving patients without follow-up support. Additionally, the healthcare ecosystem lacks a **centralized and unified platform** where doctors and patients can efficiently manage appointments, medical records, and communication in a secure environment.

These challenges motivated the development of Book My Doctor — a comprehensive, user-friendly, and secure platform that addresses these gaps by integrating **online booking, telemedicine, chatbot support, home visits,** and **digital health management** into a single system.

1.3 Aim and Objective

1.3.1 Aim

To build a comprehensive web-based Doctor-Patient platform that enables seamless online appointment booking, enhances real-time communication between doctors and patients, integrates telemedicine capabilities, and provides a secure, user-friendly, and accessible digital healthcare experience for urban and rural users alike, ultimately reducing wait times, improving healthcare equity, and promoting preventive care through digital tools.

1.3.2 Objectives

The primary objectives of this project are to create an efficient, inclusive, and technologically advanced healthcare platform that addresses key challenges in modern medical access. This includes bridging gaps in traditional systems by leveraging web technologies for better patient outcomes and doctor efficiency.

- To develop an online doctor appointment system for easy scheduling, with real-time availability checks and conflict-free booking to minimize errors and delays.
- To provide separate dashboards for doctors and patients for better management, including analytics for doctors and personalized health trackers for patients.
- To integrate chatbot support for assistance and FAQs, powered by AI to offer 24/7 guidance on symptoms, bookings, and general health queries.
- To add a home visit booking option for patients, especially targeting elderly, disabled, and chronically ill individuals to enhance accessibility.
- To ensure data privacy and security for all users, complying with standards like HIPAA or GDPR through encryption and access controls.
- To design a simple, responsive, and intuitive user interface that works across devices, ensuring usability for all age groups and tech literacy levels.
- To enable search filters by specialty, location, and doctor availability, along with ratings and reviews for informed decision-making.
- To maintain digital patient history for future reference, including e-prescriptions, lab results, and consultation notes in a centralized repository.
- To provide notifications and reminders for upcoming appointments via email, SMS, and app pushes to reduce no-shows and improve adherence.

- To improve doctor-patient communication via messages or updates, including post-consultation follow-ups, teleconsultation integration, and shared health updates.

1.4 Report Outline

The following outline provides a comprehensive overview of the chapters and sections included in this report, illustrating the scope, features, methodology, and implementation details of the developed Book My Doctor system. It guides the reader through the project's background, analysis, design, and outcomes, ensuring a logical flow from conception to evaluation.

- **Abstract**: Summarizes the project's purpose, key features, technologies used, and benefits in a concise manner.
- **Introduction**: Discusses the background of healthcare digitization, motivation for the platform, aims, objectives, and overall scope of the work.
- **Literature Survey**: Reviews existing research, similar platforms, and technologies in online healthcare systems, highlighting gaps and innovations.
- **Existing System**: Analyzes current appointment booking and telemedicine solutions, their features, limitations, and drawbacks.
- **Problem Definition**: Clearly defines the challenges in traditional and digital healthcare access, such as accessibility issues, security concerns, and inefficiencies.
- **Proposed System**: Describes the Book My Doctor platform in detail, including enhanced features like AI chatbot, home visits, and digital records to overcome identified problems.

- **System Architecture**: Explains the technical design, including front-end, back-end, database structure, security measures, and integration of modules like telemedicine.
- **References**: Lists all sources, papers, websites, and tools referenced throughout the report for further reading and validation.

Chapter 2

Study of The System

2.1 Literature Survey

2.1.1 Online Doctor Appointment System using Web Technology - 2022 [1]

Authors: Venkatesh Rallapalli, Dipti Menghani, Hema Gallani, Dr. Dashrath Mane

Idea:

This paper proposed a web-based application to streamline hospital appointment scheduling using cloud infrastructure. It aimed to reduce patient wait times, automate scheduling, and enhance workflow with real-time updates and centralized slot management. Patients can register, select doctors, and get email/SMS confirmations, while admins monitor occupancy. Basic security with authentication and encryption supports contactless post-COVID operations.

Technology/Tools:

- **Frontend:** HTML, CSS, JavaScript, Bootstrap – for responsive, mobile-friendly interfaces.
- **Backend:** PHP/Java/Python – for handling requests, authentication, and processing.
- **Database:** MySQL/PostgreSQL – for storing user data and appointments efficiently.

- **Cloud:** AWS/Google Cloud – for scalable hosting, load balancing, and secure storage.

Observation & Relevance:

The system effectively reduces administrative overhead by up to 40% in simulations but lacks patient-centric features like AI chatbots, teleconsultation, home visit booking, and separate dashboards for doctors and patients. It also misses multi-factor authentication and accessibility options like voice commands. These gaps highlight the need for comprehensive platforms addressing rural access and regulatory compliance. Our Book My Doctor enhances this with AI-powered chatbots, teleconsultation, GPS-enabled home visits, role-based dashboards, and secure record sharing, creating a holistic digital health ecosystem.

2.1.2 Telemedicine and Health Policy: A Review - 2020 [2]

Authors: Clemens Scot Kruise, Kelly Williams, John Bohls

Idea:

This study reviewed global telemedicine platforms, focusing on their role in improving healthcare access during the COVID-19 pandemic. It highlighted online consultations and electronic health records (EHRs) for remote care, reducing hospital visits and supporting rural patients through policy changes like relaxed regulations.

Technology/Tools:

As a literature review, the paper collated findings from various telemedicine platforms without specifying a single implementation. Common technologies identified include:

- **Cloud-Based Systems:** Platforms hosted on AWS, Azure, or Google Cloud for scalable, secure storage and access to patient

data across regions.

- **Video Conferencing Tools:** Solutions like Zoom, Webex, or proprietary APIs for real-time doctor-patient consultations, ensuring HIPAA-compliant encryption.
- **Mobile Apps:** Patient-facing applications (iOS/Android) for scheduling, reminders, and accessing consultation summaries on the go.
- **Electronic Health Records (EHRs):** Systems like Epic or Cerner for storing medical histories, prescriptions, and lab results, enabling seamless data sharing between providers.

Observation & Relevance:

Telemedicine boosts accessibility, cutting in-person visits and aiding rural care. However, platforms lack features like appointment scheduling, AI assistance, or full EHR integration. Our Book My Doctor addresses this with booking, AI chatbots, home visits, and secure EHRs, enhancing patient-centric care.

2.1.3 A Smart Healthcare System For Patient Monitoring using IOT(IEEE) - 2022 [3]

Authors: P Rajan Jeyaraj, S. Edward Rajan

Idea:

This research introduced a smart healthcare system for real-time patient monitoring via IoT sensors, collecting vital signs like heart rate and temperature. Data is displayed on a doctor's web dashboard, enabling timely interventions and remote tracking, especially for chronic or critical patients, reducing hospital visits and enhancing care efficiency.

Technology/Tools:

- **Hardware:** IoT sensors (e.g., heart rate, temperature) with microcontrollers like Arduino/NodeMCU for data capture and transmission.
- **Backend:** Cloud storage (e.g., AWS IoT Core) for real-time data aggregation; web dashboards for doctors to visualize patient metrics.

Observation & Relevance:

The system improves care delivery through continuous monitoring, cutting response times for critical cases. However, it lacks online appointment scheduling, direct patient-doctor communication, and AI-driven diagnostic support. Our Book My Doctor integrates these with IoT monitoring, adding booking, AI chatbots, and secure record sharing for a comprehensive healthcare platform.

2.1.4 Conversational Chatbots in Healthcare - 2022 [4]

Authors: Dr Vinod Kimbahune, Prof. Dhanashri Patil, Sakshi Nimbalkar, Suyash Alekar, Meghraj Yadav, Snehal Desai

Idea:

This paper proposes an AI-based medical chatbot for diagnosing illnesses and providing basic health information from user symptoms, acting as a preliminary consultant to reduce costs and improve access before doctor visits. It uses natural language processing to extract keywords, match queries via similarity scores, and suggest remedies or appointments, enabling remote interactions via messaging platforms.

Technology/Tools:

- **NLP:** For intent recognition, NLU/NLG, and processing user inputs with n-gram, TF-IDF, cosine similarity for query matching.

- **ML Algorithms:** SVM for classification/hyperplane creation; KNN for similarity-based categorization; applied in Python on datasets like cardiac data.
- **Frameworks:** RASA for conversational development; Dialogflow for integration; Ngrok for localhost-to-public links.
- **Backend:** Database for knowledge storage; data preprocessing (stopwords removal, analysis).

Observation & Relevance:

The chatbot achieves 65% precision, 65% recall, and 71% precision in symptom diagnosis, offering user-friendly preliminary assessments. However, it lacks advanced symptom details (location, duration), full disease coverage, and voice support, limiting it to basic tools. Our Book My Doctor builds on this with integrated appointment booking, AI chatbots for triage, secure records, and home visits, creating a comprehensive platform beyond isolated diagnosis.

2.2 Existing System

2.2.1 DOCTORIDUNIYA.COM [6]

Idea:

DoctoriDuniya is a digital healthcare platform that bridges the gap between patients and healthcare providers. It offers features such as online appointment booking, virtual consultations, 24×7 online access, multi-specialty support, e-prescriptions, and follow-ups. The platform claims to maintain a secure system with verified doctors.

Features:

- 24×7 online consultations
- Multi-specialty access; available via app and web
- Transparent fees, e-prescriptions, and follow-ups
- Verified doctors; claims of secure and private system

Drawbacks:

- Online payment not fully integrated; some regions not covered
- Still requires clinic visits/payments for in-person consultations;
- Doctor-side adoption may be limited or depends on subscription;
- Users may still have trust issues with remote consultations.

2.2.2 PRACTO.COM [7]

Idea:

Practo.com connects patients with healthcare providers, simplifying access to medical services. Users can search doctors, clinics, and hospitals based on specialty, location, and ratings, book appointments online, and consult doctors via video calls or chat.

Features:

- Search for doctors, clinics, and hospitals based on specialty, location, and ratings.
- Book appointments online.
- Consult doctors via online video calls or chat.

Drawbacks:

- **Limited Coverage in Rural Areas:** Practo's network is strongest in urban regions; rural areas may lack sufficient healthcare providers, limiting accessibility for those users.
- **Home Visit:** Practo does not offer doctor home visit appointments for elderly or mobility-impaired patients unable to visit clinics.

This platform streamlines urban healthcare access but falls short in rural coverage and home visit support, areas where Book My Doctor enhances accessibility with broader reach and home visit options.

2.2.3 LYBRATE.COM [8]

Idea:

Lybrate is an online healthcare platform designed to make medical consultation more accessible, convenient, and transparent for patients across India. The platform not only facilitates online appointment booking for in-person consultations but also promotes preventive healthcare by offering daily health tips, expert articles, and wellness advice.

Features:

- Chat and video consultations with verified doctors
- Online appointment booking for clinics and hospitals;
- Provides a vast database of doctors from multiple specialties;

- Digital health records storage and easy access for patients;

Drawbacks:

- No dedicated home visit option for elderly or disabled patients;
- Paid subscription is required for some premium or priority features;
- Internet connectivity issues can affect video consultation quality;
- Some users report inconsistent response times from doctors
- Trust and privacy concerns among first-time users of telemedicine platforms.

2.3 Table of Comparison

Table 2.1: Comparison of Related Work in Online Healthcare Systems

Paper Title	Year	Advantages	Limitations
Online Doctor Appointment System using Web Technology	2022	Efficient online scheduling and real-time updates; Reduces waiting time and admin workload; Supports cloud-based data management and contactless operations.	Lacks AI chatbot and teleconsultation; No home visit feature; Limited security (no multi-factor authentication); No separate dashboards for patients/doctors.
Telemedicine and Health Policy: A Review	2020	Enhances remote healthcare access; Supports online consultation and EHRs; Useful for rural and pandemic scenarios.	Lacks integrated appointment system; No AI support or chatbot feature; Incomplete EHR and diagnostics integration.
A Smart Healthcare System for Patient Monitoring using IoT (IEEE)	2022	Enables continuous remote health monitoring; Real-time data via IoT sensors; Helps in early detection for chronic patients.	Does not support appointment booking or chat; No AI-driven analysis; Lacks patient-doctor communication features.
Conversational Chatbots in Healthcare	2022	Provides instant responses for basic health queries; Reduces preliminary consultation costs; Easy to use via chat platforms.	Limited disease coverage; No voice input or emotional analysis; Not integrated with doctor booking or EHR.

Our **Book My Doctor** system addresses these limitations by offering home visit bookings, enhanced rural accessibility, and a free-tier model with AI chatbot support, improving inclusivity and convenience.

Table 2.2: Comparison of Existing Online Healthcare Platforms

Platform	Advantages	Limitations
DOCTORI DUNIYA .COM	• 24×7 online consultation access • Multi-specialty support via single platform • Verified and trusted doctors • Secure data management and e-prescriptions	• Limited online payment integration • Partial regional/rural coverage • Some consultations still require clinic visits • Dependent on doctor subscription model
PRACTO .COM	• Simplifies doctor search and appointment booking • Supports online consultation via video/chat • Verified listings and user reviews • Strong urban presence	• Weak rural connectivity and access • No home visit facility for elderly • Requires stable internet connection • Less effective for non-tech-savvy users
LYBRATE .COM	• Offers chat/video consultations with verified doctors • Digital health record management • Daily health tips and preventive care features • Promotes patient-doctor communication	• No dedicated home visit support • Paid subscription for premium features • Internet issues may affect consultations • Trust and privacy concerns among new users

Chapter 3

Proposed System

3.1 Problem Statement

In the current healthcare system, patients face **long waiting times, manual scheduling, and limited access to doctors,** especially in remote or rural areas. Elderly and disabled patients encounter additional barriers due to mobility issues, necessitating home-based consultation options. Additionally, the lack of continuous communication between doctors and patients outside clinic visits hinders effective care and follow-up.

To address these challenges, the Book My Doctor system proposes a web-based platform that:

- Streamlines doctor-patient interaction through intuitive interfaces and messaging.
- Enables online and home visit appointment booking for flexible access.
- Integrates chatbot assistance for instant guidance and FAQs.
- Maintains secure digital medical records with encrypted storage.
- Provides timely notifications and reminders to enhance adherence.

This ensures a fast, accessible, and reliable healthcare experience for patients while supporting doctors and administrators with efficient digital tools.

3.2 Scope

The **Book My Doctor** system is a multi-user healthcare platform designed to streamline doctor–patient interactions through an integrated online environment. It primarily serves three user groups—patients, doctors, and administrators—each having distinct roles and privileges.

- **Patients:**

  Patients can book online or home visit appointments, access e-prescriptions, and manage personal health records through a secure, user-friendly interface.

- **Doctors:**

  Doctors can manage schedules and appointments, maintain digital patient records, and view consultation histories through a streamlined, secure platform.

- **Administrators:**

  Administrators can oversee database records, monitor user activity, and maintain system integrity and security through robust management tools.

Scope Highlights:

- Web-based platform accessible via browsers and mobile devices.
- Chatbot integration for patient assistance.
- Secure data storage using encryption.
- Notification and reminder system for appointments.
- Future extensions include online video consultation and AI-based diagnosis support.

3.3 Hardware Requirements

Component	Specification
Processor	Intel Core i3 or above
RAM	Minimum 4 GB
Hard Disk	Minimum 250 GB
Display	1024 × 768 resolution or higher
Internet	Stable broadband connection
Devices	Desktop/Laptop (for doctor and admin), Smartphone (for patient access)

3.4 Software Requirements

Category	Specification
Operating System	Windows 10 / Linux / macOS
Frontend	HTML, CSS, JavaScript
Backend	Python Flask (v3.13.7)
Database	MySQL (v8.4 LTS)/Firebase
Server	XAMPP / Flask Localhost Server
IDE/Editor	VS Code / PyCharm
Other Tools	Chatbot API (Python-based), Notification Scheduler, Encryption Libraries

Chapter 4

Design of the System

4.1 Approach

The proposed approach for the healthcare system focuses on delivering an intuitive, role-based platform for patients and doctors using a modular web-based architecture. The system integrates multiple modules—user navigation, role authentication, patient dashboard, doctor dashboard, and logout—to ensure seamless interaction and efficient healthcare services. The primary goal is to provide real-time access to functionalities like appointment booking, doctor search, chatbot assistance, and record management, all running on standard web browsers without specialized hardware. By leveraging HTML, CSS, and JavaScript for the front-end, Python Flask for back-end logic, and a database (e.g., Firebase) for data management, the system ensures accessibility, scalability, and security. This modular design enables robust, real-time performance for role-specific user interactions.

4.1.1 Flow Chart

The flowchart maps the system's workflow to code modules, starting from the homepage to role selection (patient or doctor). Patient modules include dashboard, chatbot, search, booking, and confirmation. Doctor modules cover dashboard, appointment management, and record updates. Both paths end at logout, guiding efficient implementation of role-based features.

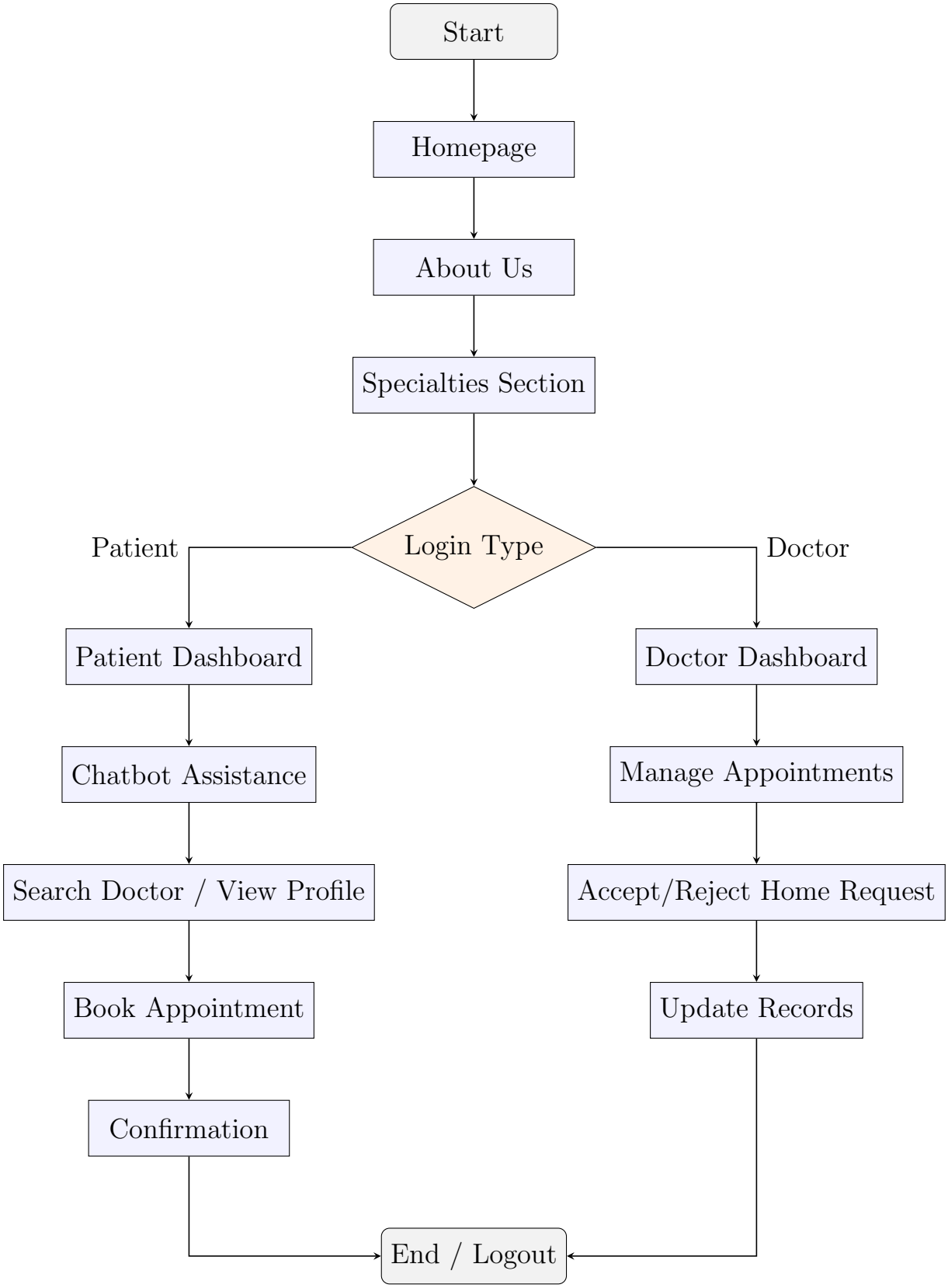

Figure 4.1: Flowchart

4.2 Technology Stack

4.2.1 Tools Required

Front End

The front-end is responsible for the user interface and user experience, providing an intuitive and responsive design for patients, doctors, and administrators.

- **Design Tool:** Figma – used for creating interactive prototypes and visual layouts.
- **HTML5:** Provides the structure of web pages and ensures semantic and accessible content.
- **CSS3:** Used for styling, layouts, animations, and responsive design for multiple devices.
- **JavaScript (ES6):** Adds interactivity, dynamic content, and client-side validation.

Back End

The back-end handles the application logic, request processing, and server-side operations.

- **Python Flask (v3.13.7)**: Lightweight web framework for building APIs, handling user authentication, appointment management, and server-side logic.
- **Firebase:** Cloud-based real-time database used to store patient data, doctor profiles, appointments, and chat history.

4.3 System Architecture

The Book My Doctor system adopts a three-tier architecture to ensure scalability, security, and seamless user interaction. The **Presentation Layer** delivers a responsive frontend (HTML5, CSS3, JavaScript) for patients, doctors, and administrators via browsers or mobile devices. The **Application Layer**, powered by Python Flask, handles core logic, including appointment booking, chatbot assistance, and notifications. The **Data Layer** uses MySQL for secure storage of user profiles, health records, and appointment data, with encryption for privacy.

Key components include:

- **User Authentication:** Secure login for patients, doctors, and admins.
- **Booking Module:** Real-time scheduling with home visit options.
- **Chatbot Service:** AI-driven assistance for FAQs and booking support.
- **Notification System:** Email/SMS reminders for appointments.
- **Records Management:** Digital storage of e-prescriptions and health history.

4.3.1 Block Diagram

The block diagram illustrates the system's workflow, with users interacting through the frontend, processed by backend services, and stored in the database.

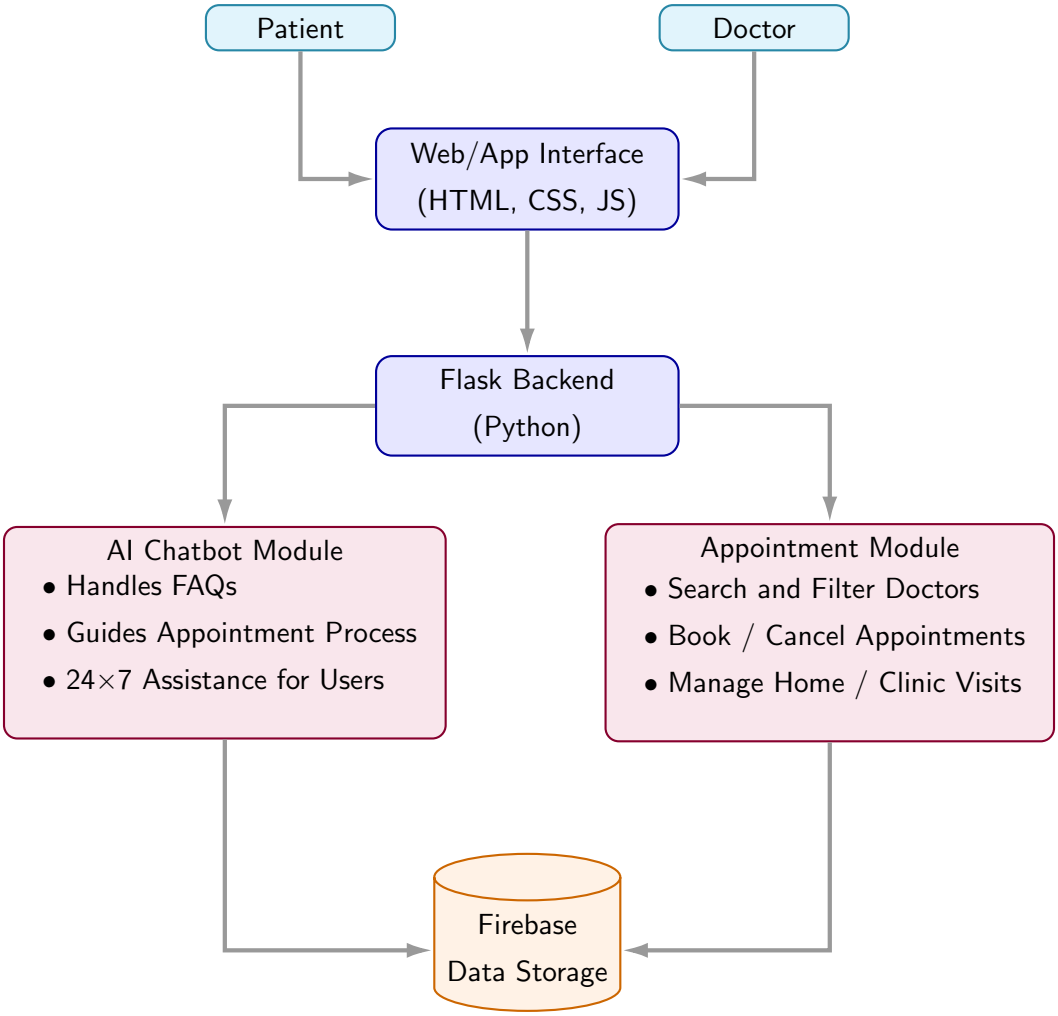

Figure 4.2: System Architecture

This architecture supports efficient doctor-patient interactions, with modular design for future enhancements like video consultations and IoT integration.

Chapter 5

Conclusion and Future Scope

5.1 Conclusion

The **Book My Doctor** system tackles key healthcare challenges, such as long wait times, limited rural access, poor doctor-patient communication, and inefficient record management. By integrating appointment booking, home visit requests, AI chatbot assistance, notifications, and secure medical history management into a web-based platform, it achieves:

- **Enhanced Patient Convenience:** Online booking, medical history tracking, and home visit requests reduce clinic visits.
- **Efficient Doctor Management:** Digital schedules, and records cut administrative workload and improve service quality.
- **Real-Time Interaction:** Notifications and chatbot support ensure timely patient guidance.
- **Data Security and Privacy:** Encryption safeguards patient data, meeting privacy standards.
- **Responsive Interface:** Accessible across desktops, laptops, and smartphones for seamless user experience.

This platform bridges gaps between patients and providers, offering faster, accessible, and reliable healthcare while reducing doctor workload and boosting patient satisfaction.

5.2 Future Scope

The Book My Doctor system can evolve into a more intelligent and comprehensive platform with the following enhancements:

- **AI-Based Diagnosis:** Machine learning for symptom analysis and predictive health alerts.
- **Mobile Apps:** Native Android/iOS apps for greater accessibility.
- **Payment Gateway:** Secure payment integration for consultations and prescriptions.
- **Advanced Analytics:** Dashboards for patient trends.
- **Multilingual Support:** Language options for diverse populations.
- **Smart Reminders:** Automated follow-ups and health tips to improve adherence.

Impact of Future Enhancements:

These upgrades will create a scalable, holistic platform, improving patient outcomes, doctor efficiency, and the digital healthcare ecosystem, aligning with telemedicine growth trends.

Acknowledgement

Success of a project like this involving high technical expertise, patience, and massive support of guides, is possible when team members work to-gather. We take this opportunity to express our gratitude to those who have been instrumental in the successful completion of this project. We would like to appreciate the constant interest and support of our mentor guide **Mrs. Bhakti Aher** in our project and aiding us in developing a flair for the field of Your Domain. We would always cherish the journey of transforming the idea of our project into reality. We would like to show our appreciation to **Mrs. Bhakti Aher** for her tremendous support and help, without whom this project would have reached nowhere. We would also like to thank our project coordinator **Mrs. Chetana Badgujar** for providing us with regular inputs about documentation and project timeline. A big thanks to our **HOD Dr. M. Kiruthika** for all the encouragement given to our team. We would also like to thank our principal, **Dr. S. M. Khot**, and our college, **Fr.C.Rodrigues Institute of Technology, Vashi**, for giving us the opportunity and the environment to learn and grow.

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[12] VS Code – Integrated Development Environment, https://code.visualstudio.com

.1 Appendix A: Timeline Chart

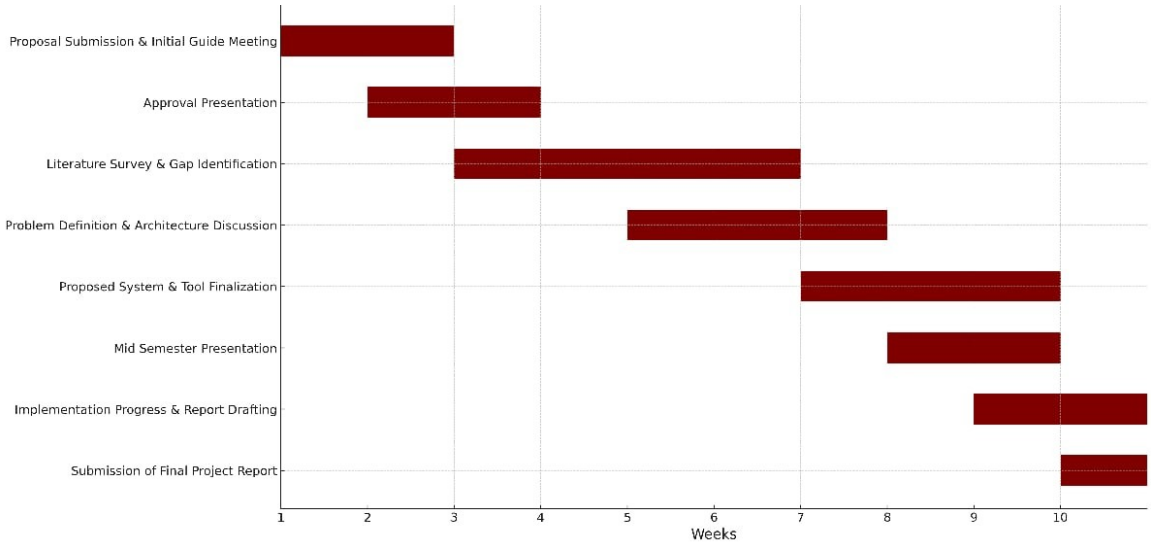

Figure 1: Timeline Chart